

■ OFFICE CHAOS

The Workplace Stress-Relief Multiplayer Game

Product Requirements Document · Agile Project Plan

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1. Executive Summary

Office Chaos is a browser-based, real-time multiplayer game designed exclusively for use among colleagues. The core premise is simple: when a coworker is driving you mad, instead of snapping at them in real life, you log into Office Chaos and virtually annihilate their desk, set their meeting room on fire, slap them across the open-plan floor, or unleash a swarm of rubber ducks on their workstation.

The game is not a productivity tool — it is a social, cathartic stress-relief experience that lives alongside the real workplace. It is funny by design. It is competitive by design. And it only makes sense when played with the people you actually work with.

This document covers the full Product Requirements (functional and non-functional) and the Agile delivery plan across six sprints targeting a working MVP in approximately 12 weeks.

2. Goals & Success Metrics

2.1 Product Goals

- Provide a safe, humorous outlet for real workplace frustration without any real-world conflict.
- Create a social layer that makes colleagues laugh together, not at each other.
- Keep onboarding under 2 minutes so anyone in the office can join a session instantly.
- Be the game people open on a Monday morning after a brutal weekend of missed Slack pings.

2.2 Success Metrics

Metric	Target (3 months post-launch)	How Measured
Weekly Active Users (WAU)	≥ 60% of registered users	Auth logs
Session length	≥ 8 minutes average	Client telemetry
Colleague invites sent	≥ 2 per new user	Invite service
Leaderboard views per week	≥ 3 per active user	Analytics
Stress meter filled (meeting trigger)	≥ 40% of sessions	Game event log

3. User Personas

Persona A – The Venter

Name	Marco, 31, Backend Engineer
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Frustration	Three back-to-back stand-ups before 10 AM every Monday
Goal	Quick, satisfying destruction with minimal setup
Behaviour	Logs in, targets the PM who scheduled all three meetings, burns desk, logs out

Persona B – The Competitor

Name	Sara, 27, Growth Analyst
Frustration	Colleague always tops the OKR board; wants any competitive outlet
Goal	Climb the Weekly Leaderboard, brag in Slack
Behaviour	Plays every day, optimises combo chains, screenshots leaderboard rank

Persona C – The Social Instigator

Name	Leo, 34, Engineering Manager
Frustration	Team morale dips after sprint retros
Goal	Get the whole team playing together as a bonding moment
Behaviour	Creates a team workspace, invites everyone, posts results in the team channel

4. Feature Requirements

Requirements are tagged with a priority: **P0** = must-have for MVP, **P1** = important for V1, **P2** = future enhancements.

FR-1 Authentication & Workspace

ID	Priority	Requirement	Notes
FR-1.1	P0	Users can register with email + password	bcrypt hashing
FR-1.2	P0	Create or join a named Workspace (company/team); workspaces are isolated game rooms	Invite-based isolation
FR-1.3	P0	Invite colleagues via shareable link	Link expires after 48 h
FR-1.4	P1	Google OAuth login	Single click for GSuite workplaces

ID	Priori ty	Requirement	Notes
FR-1.5	P1	Workspace admin can remove users	Moderation safety valve

FR-2 Avatar Creator

ID	Priori ty	Requirement	Notes
FR-2.1	P0	Select skin tone from a palette of 6 options	Fitzpatrick scale reference
FR-2.2	P0	Select hairstyle from ≥ 8 preset silhouettes	Illustrated SVG assets
FR-2.3	P0	Select hair colour from a 12-colour palette	Includes fantasy colours
FR-2.4	P0	Select outfit from ≥ 6 office archetypes (casual, suit, hoodie, etc.)	Affects in-game sprite
FR-2.5	P0	Avatar displayed to all players in the same workspace	Stored as JSON config
FR-2.6	P1	Display name + job title tagline	Shows on hover tooltip
FR-2.7	P2	Accessory slot: glasses, headphones, coffee cup	Unlockable via XP

FR-3 Stress Meter & Meeting Trigger

ID	Priori ty	Requirement	Notes
FR-3.1	P0	Each player has a personal Stress Meter (0–100)	Visible on HUD
FR-3.2	P0	Stress increases passively over time during a session (+1/min)	Simulates work week
FR-3.3	P0	User can log a 'meeting' event; each adds +10 stress	Manual trigger button in UI
FR-3.4	P1	Optional Google Calendar integration to auto-detect meetings and raise stress	Needs OAuth scope
FR-3.5	P0	At 100% stress the player enters RAGE MODE: +50% speed, 2x damage for 30 s	Core fun mechanic
FR-3.6	P0	Successful destruction actions reduce stress (-5 per action)	Reward loop
FR-3.7	P0	Stress meter is visible to all teammates	Broadcast via WebSocket

FR-4 Game World & Isometric Engine

ID	Priori ty	Requirement	Notes
FR-4.1	P0	Game world rendered in 2D isometric view in the browser	Phaser 3 + isometric plugin
FR-4.2	P0	Office floor plan: open desks, meeting rooms, kitchen, server room	Tilemap-based layout

ID	Priori ty	Requirement	Notes
FR-4.3	P0	Each player's desk labelled with their avatar name	Personalises targets
FR-4.4	P0	Players can move with arrow keys / WASD	Smooth tile-based movement
FR-4.5	P0	Real-time positions of all online players are visible	WebSocket broadcast
FR-4.6	P1	Office layout configurable per workspace (drag-drop room editor)	Personalisation hook

FR-5 Destruction & Prank Mechanics

ID	Priori ty	Requirement	Notes
FR-5.1	P0	Target an unoccupied desk and trigger a destruction animation (coffee spill → papers fly → chair topple)	3-frame sequence
FR-5.2	P0	Set fire to a room (particle effect, spreading); extinguisher item to stop it	Core chaos tool
FR-5.3	P0	Destruction actions only available when target player is offline / AFK	Cat-and-mouse tension
FR-5.4	P0	Target player sees a comedic 'victim notification' on next login	Toast + animation
FR-5.5	P1	Combo system: chaining ≥ 3 actions within 10 s gives a score multiplier	Skill ceiling mechanic
FR-5.6	P1	Prank toolbox expands with XP: post-it bomb, rubber duck flood, printer jam	Unlockable at levels 5/10/15
FR-5.7	P0	Player cannot target themselves	Server-side guard

FR-6 PvP Combat

ID	Priori ty	Requirement	Notes
FR-6.1	P0	A player can click on a nearby online player to initiate a punch/slap attack	Proximity-gated: must be within 2 tiles
FR-6.2	P0	Combat is server-authoritative: attack validity and hit detection resolved on the server	No client-side hit registration
FR-6.3	P0	Each player has a HP bar (e.g. 3 hit points); depleted HP triggers a knockdown	HP bar visible on HUD
FR-6.4	P0	Knockdown plays a comical full-body collapse animation on the defeated player's sprite	Stars / birds circling effect
FR-6.5	P0	Defeated player respawns at their own desk after 5 seconds with full HP	Respawn timer shown on screen
FR-6.6	P0	Attacker cannot hit the same player again for 2 seconds after a hit (cooldown)	Prevents click-spam

ID	Priori ty	Requirement	Notes
FR-6.7	P0	Players in RAGE MODE deal 2x punch damage (1 hit KO)	Integrates with stress mechanic
FR-6.8	P0	Combat is only possible between online players; AFK or offline players cannot be attacked	Guards against griefing
FR-6.9	P1	Hit landed triggers a comedic sound effect and floating damage label ('BONK!', 'SLAP!')	Visual + audio feedback
FR-6.10	P1	Spectators see the fight in real time; hit events broadcast to all room members	Social entertainment layer
FR-6.11	P1	A 'wanted' indicator appears above a player who has KO'd 3+ opponents in a session	Adds chase-me dynamic
FR-6.12	P2	Throwable office objects (stapler, mug, chair) as ranged attacks unlocked via XP	Post-MVP escalation

FR-7 Weekly Leaderboard

ID	Priori ty	Requirement	Notes
FR-7.1	P0	Leaderboard scoped to a Workspace; resets every Monday 00:00 UTC	Weekly competitive cycle
FR-7.2	P0	Destruction score from: destructions, combos, Rage Mode activations, pranks (weighted)	Point system
FR-7.3	P0	Separate PvP ranking: tracks KOs landed, KO streaks, and times knocked out (the shame column)	Independent from destruction score
FR-7.4	P0	Both boards visible from the main HUD at all times, togglable via tabs	Persistent side panel
FR-7.5	P0	Top 3 weekly destruction titles: Destroyer of Desks, Master of Mayhem, Fire Starter	Cosmetic reward
FR-7.6	P0	Top 3 weekly PvP titles: Office Brawler, Slap Champion, The Untouchable (most KOs, fewest deaths)	Cosmetic reward
FR-7.7	P1	All-time Hall of Shame with historical top 3 per week for both boards	Trophy room view
FR-7.8	P1	Shareable leaderboard image (auto-generated PNG via Canvas API)	For Slack/Teams posting

FR-8 Real-Time Multiplayer

ID	Priori ty	Requirement	Notes
FR-8.1	P0	Game state synchronised via WebSocket server (Socket.io)	< 100 ms latency target

ID	Priori ty	Requirement	Notes
FR-8.2	P0	Player movement broadcasted at 20 ticks/second; client-side interpolation	Smooth movement
FR-8.3	P0	Destruction, fire and combat events are server-authoritative	No client-side cheating
FR-8.4	P0	Workspace supports up to 20 concurrent players (MVP cap)	Horizontal scaling later
FR-8.5	P1	Presence indicator: online / AFK / offline per player	Critical for targeting and combat
FR-8.6	P1	In-game emoji reactions (👍 👎 🗿) during Rage Mode and KOs from spectators	Social engagement

5. Non-Functional Requirements

Category	ID	Requirement
Performance	NFR-1	Initial page load ≤ 3 s on a 10 Mbps connection
Performance	NFR-2	Game loop maintains ≥ 30 FPS on a mid-range laptop (2020+)
Scalability	NFR-3	Backend handles ≥ 50 concurrent workspace rooms without degradation
Reliability	NFR-4	Game server uptime $\geq 99.5\%$ measured monthly
Security	NFR-5	All API endpoints require valid JWT; WebSocket connections authenticated at handshake
Privacy	NFR-6	No real personal data displayed to other players without explicit consent
Compatibility	NFR-7	Supports latest 2 versions of Chrome, Firefox, Safari, Edge
Accessibility	NFR-8	Keyboard-navigable main menu; WCAG AA colour contrast on UI chrome
Data	NFR-9	Leaderboard data retained for 6 months; user data deletable on request (GDPR)

6. Proposed Tech Stack

The stack is chosen for speed of development, strong browser-game ecosystem support, and operational simplicity at small team scale.

Layer	Technology	Rationale
Game Engine	Phaser 3	Mature 2D browser game framework; isometric plugin available; large community
Frontend UI	React + Vite	HUD, menus, avatar creator, leaderboard panel sit outside the game canvas
Styling	Tailwind CSS	Rapid UI development; consistent design system
Real-time	Socket.io (WebSocket)	Battle-tested; handles reconnection, rooms, namespaces natively
Backend	Node.js + Express	JavaScript across the stack; low-latency event loop ideal for game server

Layer	Technology	Rationale
Database	PostgreSQL	Relational; strong JSON support for avatar configs and score blobs
ORM	Prisma	Type-safe schema; easy migrations
Auth	JWT + bcrypt	Stateless tokens; Google OAuth via Passport.js added at P1
Hosting	Railway / Render	Zero-ops PaaS; WebSocket support; free tier for prototyping
Asset pipeline	Aseprite → PNG sprites	Pixel/isometric art tool; exports sprite sheets directly
CI/CD	GitHub Actions	Automated test + deploy on push to main

6.1 High-Level Architecture

The architecture follows a simple client–server model with a single authoritative game server per workspace room:

- Browser client (Phaser 3 + React) communicates with the REST API for auth, avatar, and leaderboard data, and with the Socket.io server for all real-time game events.
- The Socket.io server maintains in-memory game state per workspace room (player positions, stress levels, destruction state). It persists score deltas to PostgreSQL after each significant event.
- A scheduled job (node-cron) runs every Monday at 00:00 UTC to snapshot leaderboard results and reset weekly scores.
- Static assets (sprites, tilemaps) are served via CDN (Cloudflare in front of the PaaS host).

7. Agile Project Plan

The project is organised into 6 two-week sprints, targeting a functional multiplayer MVP at the end of Sprint 5 (~week 10), with a polish and hardening sprint at week 12 before soft launch. Suggested team: 1 full-stack developer, 1 frontend/game developer, 1 part-time designer.

7.1 Epics Overview

Epic	Name	Description
E1	Foundation	Auth, workspace, DB schema, project scaffold, CI/CD
E2	Avatar Creator	Avatar builder UI, sprite rendering, persistence
E3	Game World	Isometric map, player movement, real-time sync
E4	Stress & Rage	Stress meter logic, meeting trigger, Rage Mode
E5	Destruction Toolkit	Destruction actions, fire mechanic, victim notification
E6	PvP Combat	Click-to-punch, HP bar, knockdown, respawn, combat cooldown
E7	Leaderboard	Score model, dual weekly boards (destruction + PvP), titles, reset job
E8	Polish & Launch	Performance, bug fixes, onboarding flow, soft launch

7.2 Sprint 1 – Foundation (Weeks 1–2)

Goal: The skeleton is alive. A user can register, create a workspace, and see their name on a blank isometric tile map.

Story	Epic	Description	Pts	Owner
S1-1	E1	Project scaffold: monorepo, Vite+React frontend, Express backend, Prisma + PostgreSQL	3	Full-stack
S1-2	E1	GitHub Actions CI: lint, test, deploy to staging on push	2	Full-stack
S1-3	E1	User registration and login (email/password, JWT)	3	Full-stack
S1-4	E1	Workspace create / join via invite link	3	Full-stack
S1-5	E3	Phaser 3 + isometric plugin integrated; static office tilemap renders in browser	5	Game dev
S1-6	E3	Socket.io server initialised; player can join a workspace room	3	Full-stack

Definition of Done: A registered user can log in, create a workspace, share the invite link, and both users can see a shared isometric office in their browser simultaneously.

7.3 Sprint 2 – Avatar & Movement (Weeks 3–4)

Goal: Players have a visual identity and can move around the office in real time.

Story	Epic	Description	Pts	Owner
S2-1	E2	Avatar creator UI: skin tone, hairstyle, hair colour, outfit selection	5	Frontend
S2-2	E2	Avatar config stored in DB; sprite rendered on tilemap from config	3	Game dev
S2-3	E2	Avatar and display name visible to all workspace players	2	Game dev
S2-4	E3	WASD / arrow-key movement with tile-based collision detection	5	Game dev
S2-5	E3	Real-time position broadcast via Socket.io (20 ticks/s) and client interpolation	5	Game dev
S2-6	E3	Player name label floats above sprite; desk tiles labelled with owner name	2	Game dev

Definition of Done: Multiple players from different browser tabs can see each other's custom avatars moving around the isometric office in real time.

7.4 Sprint 3 – Core Game Mechanics (Weeks 5–6)

Goal: The game is actually playable. Stress meter, destruction, fire, and victim notification all working.

Story	Epic	Description	Pts	Owner
S3-1	E4	Stress meter HUD component: fills over time (+1/min), visible to all	3	Frontend

Story	Epic	Description	Pts	Owner
S3-2	E4	'Log a meeting' button adds +10 stress; server-authoritative	2	Full-stack
S3-3	E4	Rage Mode: triggered at 100% stress; 30 s timer, 2x tool power, visual FX overlay	5	Game dev
S3-4	E5	Destruction action on unoccupied desks: 3-frame animation + debris particles	5	Game dev
S3-5	E5	Fire mechanic: fire tiles spread; extinguisher item stops it	5	Game dev
S3-6	E5	Server validates target player is offline before allowing destruction	3	Full-stack
S3-7	E5	Victim notification toast + comedic animation on next login	3	Frontend

Definition of Done: A player can fill their stress meter by logging meetings, enter Rage Mode, find an offline colleague's desk, destroy it and set fire to it. The victim sees a notification on next login.

7.5 Sprint 4 – PvP Combat (Weeks 7–8)

Goal: Players can fight each other in real time. Punching, knockdowns, respawns and the PvP ranking are all live.

Story	Epic	Description	Pts	Owner
S4-1	E6	HP bar (3 HP) per player: visible on HUD and above sprite	3	Frontend
S4-2	E6	Click-to-punch: proximity check (≤ 2 tiles), server-authoritative hit registration	5	Full-stack
S4-3	E6	2-second hit cooldown per attacker-target pair; enforced on server	2	Full-stack
S4-4	E6	Knockdown animation on defeated sprite + respawn at own desk after 5 s	4	Game dev
S4-5	E6	Rage Mode integration: 2x punch damage (1-hit KO) during active Rage Mode	2	Game dev
S4-6	E6	AFK/offline guard: combat only allowed against online, non-AFK players	2	Full-stack
S4-7	E7	PvP ranking model: KOs landed, KO streaks, times knocked out stored per session	3	Full-stack
S4-8	E6	Floating 'BONK!' / 'SLAP!' label on hit; KO event broadcast to all room members	2	Game dev

Definition of Done: Two players can click-to-punch each other, see the HP bar deplete, watch the comedic knockdown animation, and see the defeated player respawn at their desk. KO counts appear in the PvP ranking.

7.6 Sprint 5 – Leaderboard & Social Layer (Weeks 9–10)

Goal: Both competitive tracks are live and visible. The game is shareable and sticky.

Story	Epic	Description	Pts	Owner
S5-1	E7	Destruction score model: destructions, combos, Rage Mode activations → weighted points	3	Full-stack
S5-2	E7	Dual leaderboard HUD panel: destruction tab + PvP tab, live-updating	4	Frontend
S5-3	E7	Weekly reset cron job (Monday 00:00 UTC): snapshot both boards + reset scores	2	Full-stack
S5-4	E7	Destruction titles: Destroyer of Desks, Master of Mayhem, Fire Starter	2	Frontend
S5-5	E7	PvP titles: Office Brawler, Slap Champion, The Untouchable	2	Frontend
S5-6	E5	Combo system: ≥ 3 destruction actions in 10 s triggers score multiplier + banner	4	Game dev
S5-7	E3	Presence indicators: online / AFK (5 min idle) / offline shown on each desk tile	3	Full-stack
S5-8	E8	Onboarding flow: first-time modal walks new users through avatar → invite → play	3	Frontend

Definition of Done: The full gameplay loop works end-to-end. A new user can join, build an avatar, play, see their destruction score and PvP rank on the dual leaderboard, and share the workspace link.

7.7 Sprint 6 – Polish, Performance & Soft Launch (Weeks 11–12)

Goal: the game is stable, fast, and delightful enough for a real team to use. No new features — quality only.

- Performance audit: Lighthouse score ≥ 80; sprite atlas optimisation; asset lazy loading.
- WebSocket reconnection handling: automatic re-join on network drop without losing session.
- Sound effects: destruction crunch, fire crackle, punch/slap impact, Rage Mode siren (royalty-free assets).
- Playtesting session with a real team of 4–6 colleagues; fix top 5 reported issues.
- Security review: JWT expiry, rate limiting on Socket.io events, server-side combat validation audit.
- Soft launch: deploy to production, share with 1–2 willing pilot teams.

8. Future Roadmap (Post-MVP)

Release	Theme	Key Features
V1.1	Calendar Sync	Google Calendar OAuth: auto-raise stress per meeting; show today's count in HUD
V1.2	Expanded Toolkit	Unlockable pranks (post-it bomb, rubber duck flood, printer jam) gated behind XP levels 5/10/15
V1.3	Hall of Shame	All-time leaderboard trophy room; shareable weekly recap image (Canvas API → PNG)

Release	Theme	Key Features
V1.4	Office Editor	Drag-and-drop floor plan customiser per workspace; custom room names
V1.5	Emoji Reactions	Spectators send 🗡️ reactions during Rage Mode; floating emojis in the game world
V2.0	Boss Fight Mode	Weekly event: whole workspace cooperates vs an AI Boss (the CFO, the all-hands organiser)

9. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Game performance drops below 30 FPS on older hardware	Medium	High	Sprite atlas, minimal particle count, off-screen tile culling; benchmark in Sprint 1
WebSocket latency causes desync in combat (ghost hits, missed punches)	Medium	High	Server-authoritative hit resolution with client-side optimistic animation; rollback if server rejects; test with simulated 150 ms lag
WebSocket latency ruins real-time feel	Low	High	Client-side interpolation + prediction from Sprint 2; dedicated Socket.io server
Game tone misread as bullying / harassment by HR	Medium	High	Clear comedy framing on landing page; workspace admin moderation; all actions labelled as fictional; no real names required
Art asset production bottleneck	High	Medium	Use placeholder rectangle sprites in Sprints 1–3; commission final art in parallel
Low retention after first session	Medium	High	Leaderboard + weekly reset creates habitual Monday check-in; combo system provides skill depth

10. Open Questions

- Monetisation: free forever, freemium (cosmetics), or a flat team subscription? Decision needed before V1.1.
- Mobile support: should the game be playable on phone screens? The isometric view may be too small — to be validated during playtesting.
- Workspace isolation: fully private (invite-only) or a public 'stranger danger' mode for cross-company chaos?
- Progression currency: should XP be workspace-relative (resets weekly) or global (permanent account progression)?
- Sound: opt-in or on by default? A clear mute toggle is mandatory for open-plan offices.